

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-3

Faculty

Dr. Rajasekhar N V

Associate Professor G-1

SENSE, VIT-AP University

Mail: rajasekhar.venkata@vitap.ac.in

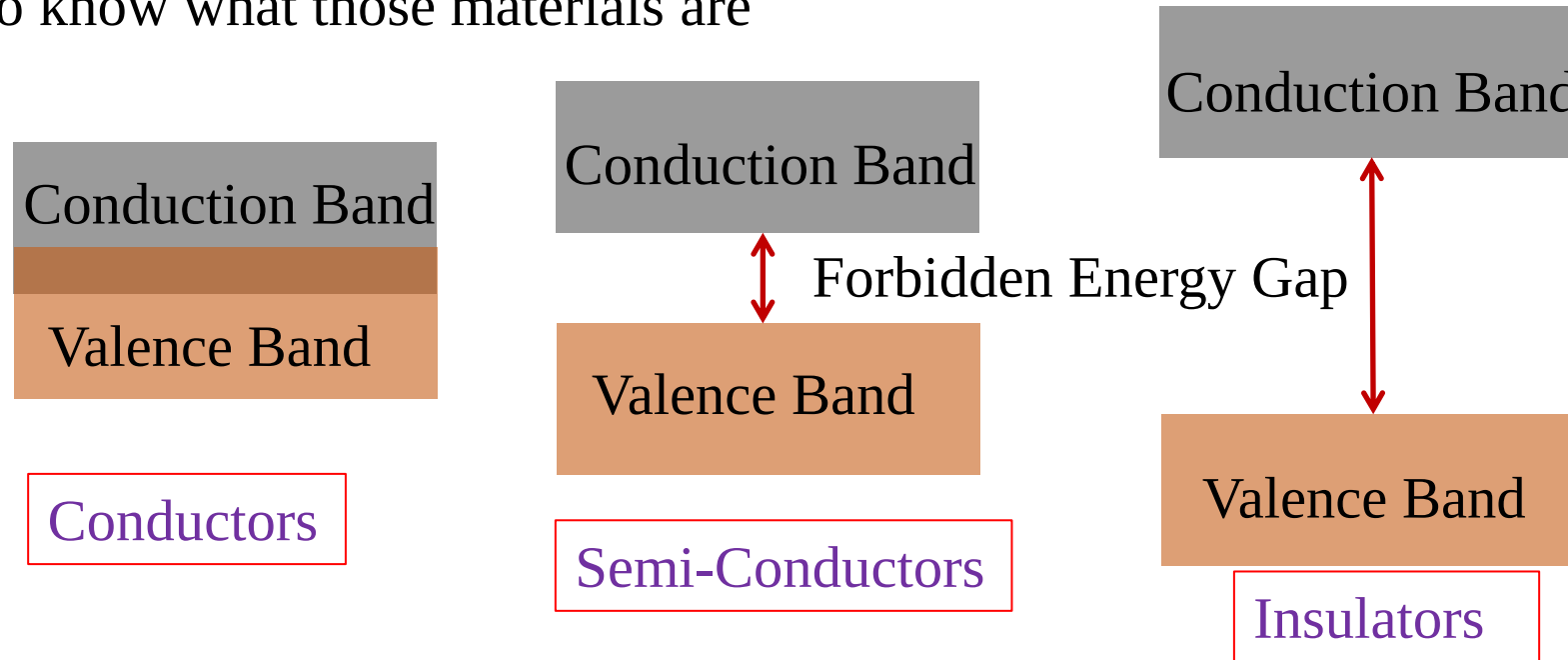
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Module-1

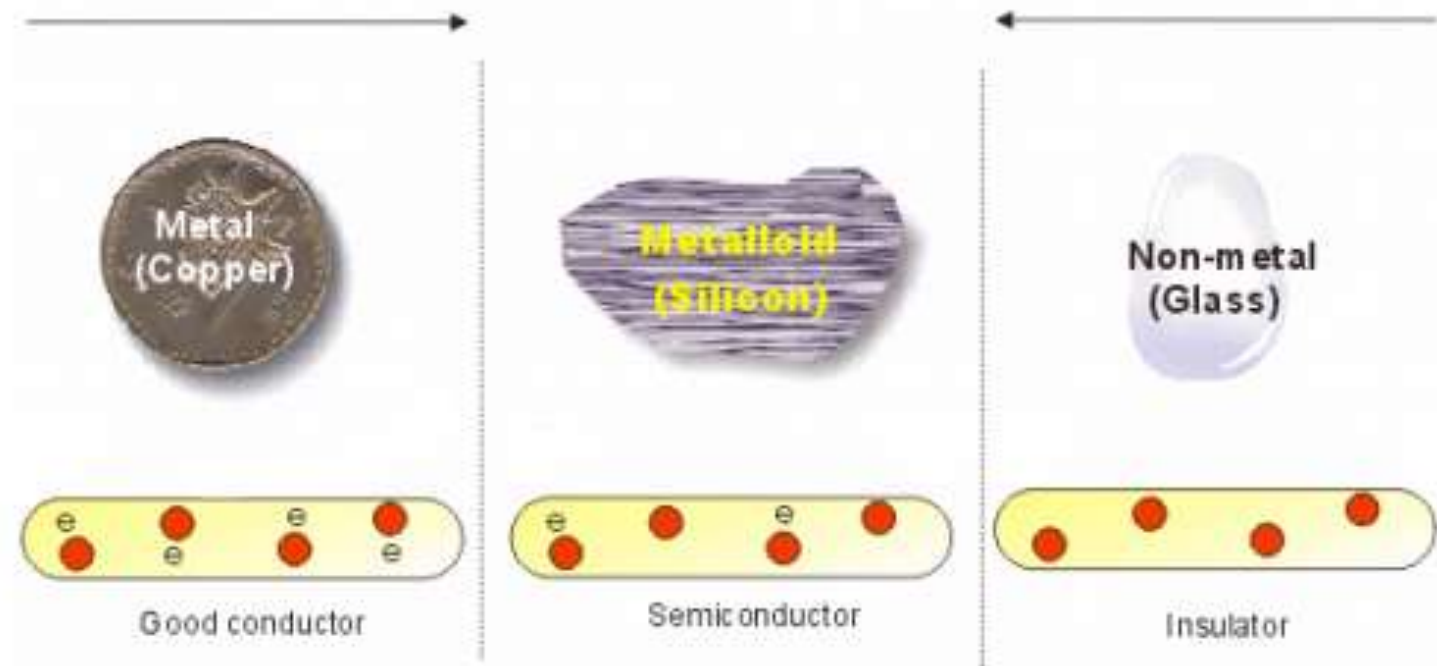
DC Circuit Analysis

Classification of Materials

- First we need to know what those materials are

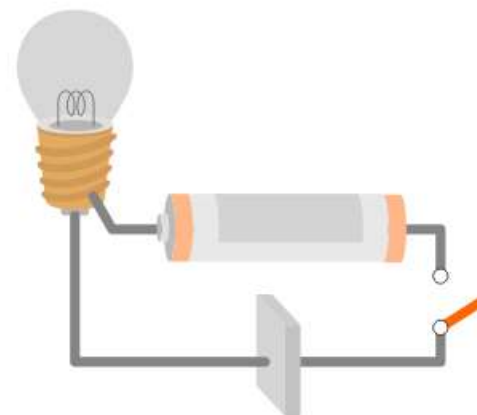
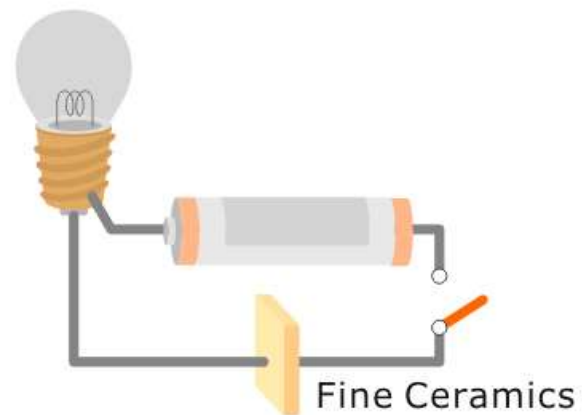


- Materials** can now be broadly **classified based on** their **conductivity** σ measured in mhos/m or Siemens/m
- Conductors are those materials with abundance of free electrons and insulators (or dielectrics) with least number



Insulating Ceramics

Metals



Electric Conductivity OF METALS



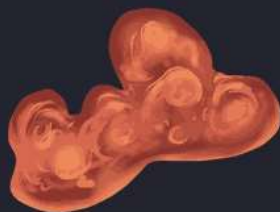
Silver



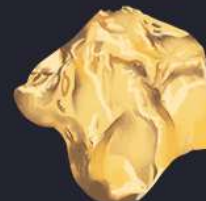
Aluminum



Iron



Copper



Gold



Electric Conductivity OF METALS



Silver

6.30×10^7



Aluminum

3.5×10^7



Iron

1.04×10^7



Copper

5.98×10^7

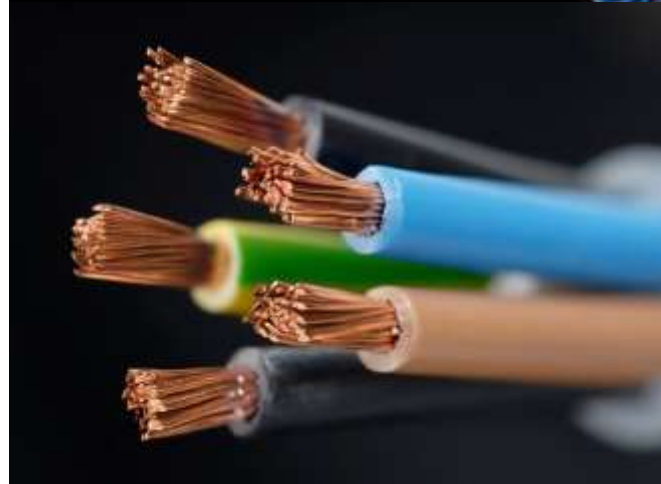
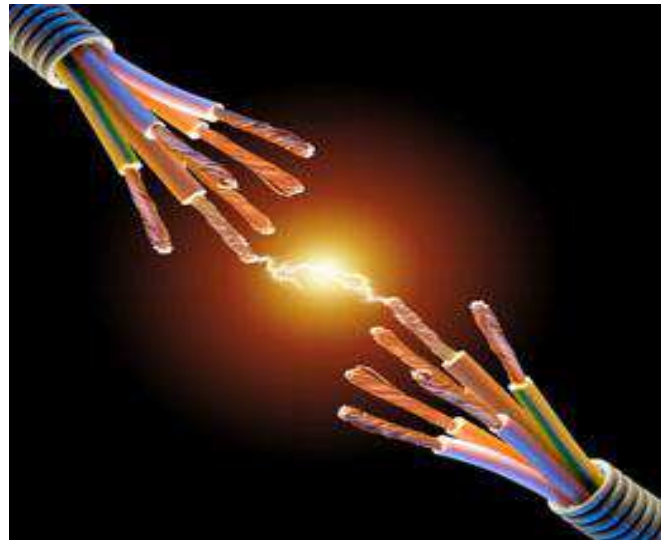


Gold

4.52×10^7

Material Vs Conductivity

Copper



Material	Conductivity (siemens/meter)
<i>Conductors</i>	
Silver	6.1×10^7
Copper (standard annealed)	5.8×10^7
Gold	4.1×10^7
Aluminum	3.5×10^7
Tungsten	1.8×10^7
Zinc	1.7×10^7
Brass	1.1×10^7
Iron (pure)	10^7
Lead	5×10^6
Mercury	10^6
Carbon	3×10^4
Water (sea)	4
<i>Semiconductors</i>	
Germanium (pure)	2.2
Silicon (pure)	4.4×10^{-4}
<i>Insulators</i>	
Water (distilled)	10^{-4}
Earth (dry)	10^{-5}
Bakelite	10^{-10}
Paper	10^{-11}
Glass	10^{-12}
Porcelain	10^{-12}
Mica	10^{-15}
Paraffin	10^{-15}

Metal



Gold

Uses



Gold ornament



Silver



Silver ornament



Mercury



Thermometer



Copper



Copper Wire



Aluminum



Aluminum ornament



Iron



Iron ornament



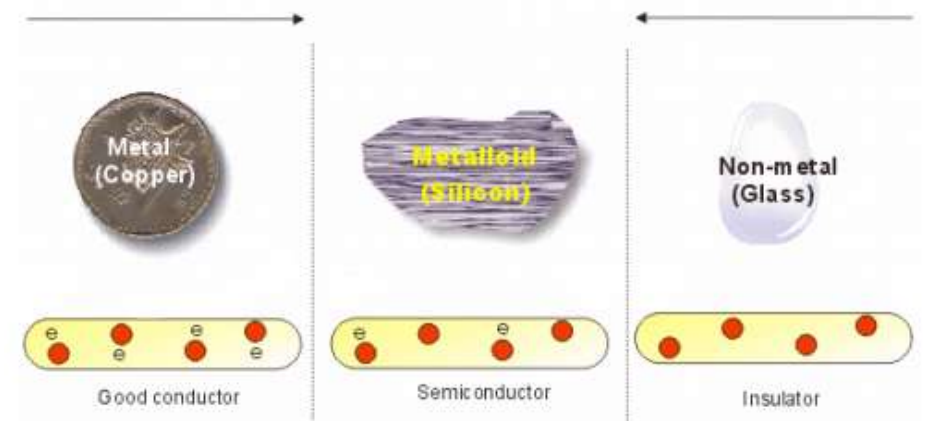
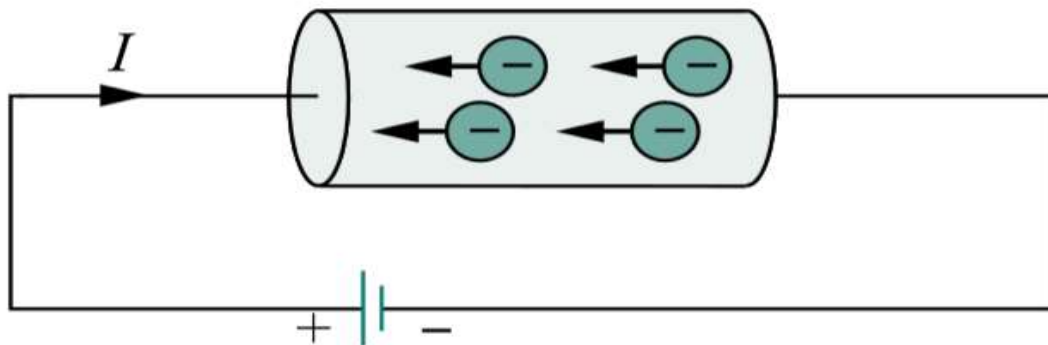
Zinc



Zinc ornament

Charge(q)

- Charge is an electrical property of the atomic particles of which consists matter, measured in coulombs(C).
- Charge on electron is negative and equal in magnitude to $1.602 \times 10^{-19} \text{ C}$
- 1 Coulomb of charge = $1 / (1.602 \times 10^{-19}) = 6.24 \times 10^{18}$ electrons



Powers of 10

Name	Common Notation	Math Notation	Exponent	Prefix
Quintillion	1 000 000 000 000 000 000	10^{18}	18	Exa (E)
Quadrillion	1 000 000 000 000 000	10^{15}	15	Peta (P)
Trillion	1 000 000 000 000	10^{12}	12	Tera (T)
Billion	1 000 000 000	10^9	9	Giga (G)
Million	1 000 000	10^6	6	Mega (M)
Thousand	1 000	10^3	3	kilo (k)
Hundred	100	10^2	2	hecto (h)
Ten	10	10^1	1	Deca (da)
One	1	10^0	0	
One Tenth	0.1	10^{-1}	-1	deci (d)
One Hundredth	0.01	10^{-2}	-2	centi (c)
One Thousandth	0.001	10^{-3}	-3	milli (m)
One Millionth	0.000 001	10^{-6}	-6	micro (μ)
One Billionth	0.000 000 001	10^{-9}	-9	nano (n)
One Trillionth	0.000 000 000 001	10^{-12}	-12	pico (p)
One Quadrillionth	0.000 000 000 000 001	10^{-15}	-15	femto (f)
One Quintillionth	0.000 000 000 000 000 001	10^{-18}	-18	atto (a)

P-1

How much charge is represented by 4,600 electrons?

Solution:

Each electron has -1.602×10^{-19} C. Hence 4,600 electrons will have
 -1.602×10^{-19} C/electron $\times$ 4,600 electrons = -7.369×10^{-16} C

P-2

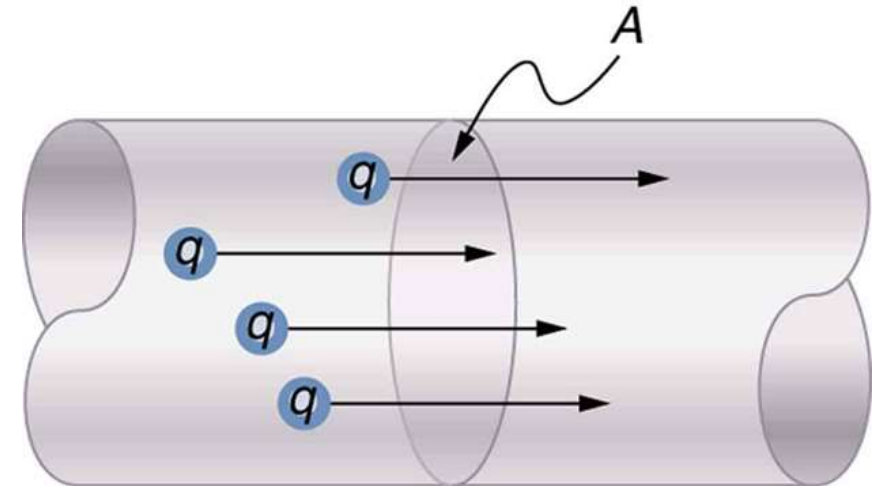
1 mC =	_____	electrons
1 μ C =	_____	electrons
1 nC =	_____	electrons

Current(i)

- Current (ampere) can be defined as the flow of electric charge (Coulombs) through a given area per unit time (seconds)

$$I = \frac{dQ}{dt}$$

Current = flow of charge



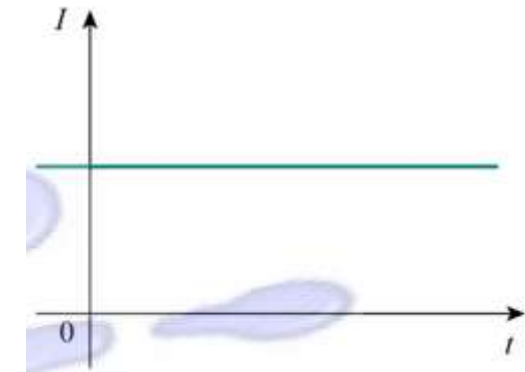
where current is measured in amperes (A), and

1 ampere = 1 coulomb/second

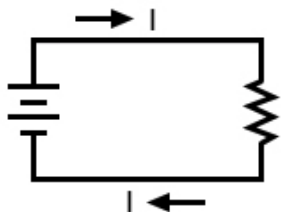
DC and AC Currents

DC Current

- A direct current(dc) is a current that remains constant with time

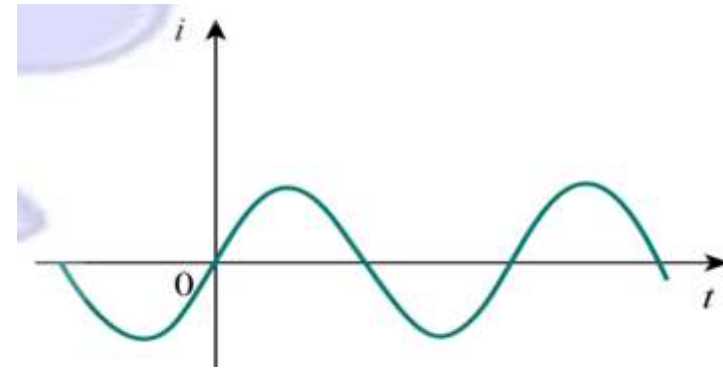


DIRECT CURRENT
(DC)

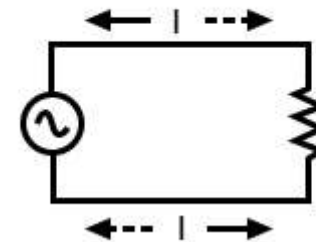


AC Current

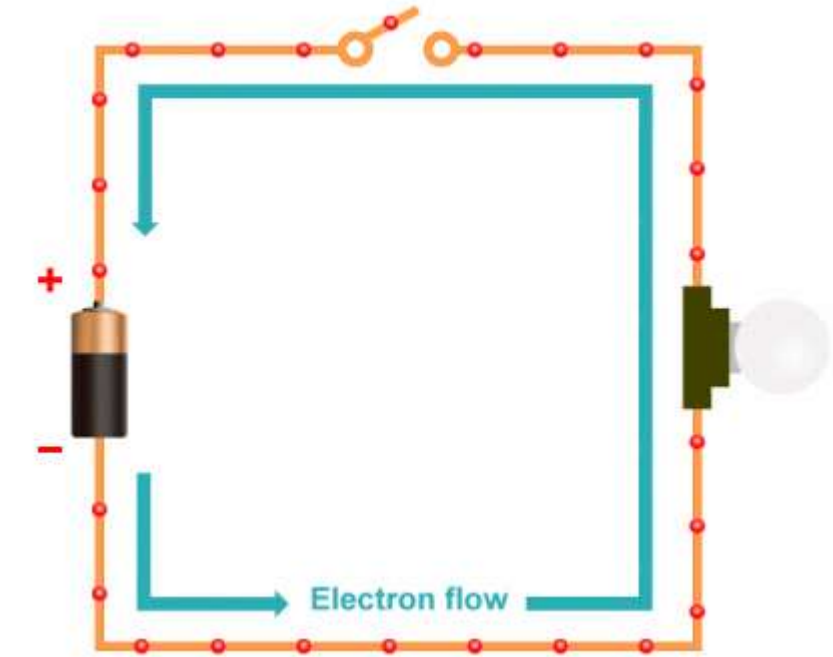
- An alternating current(ac) is a current that varies sinusoidally with time



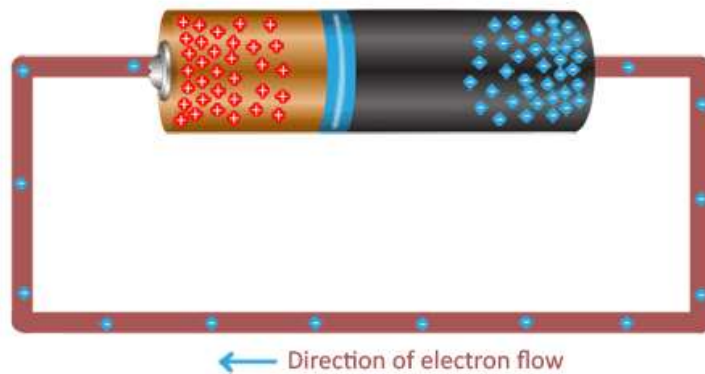
ALTERNATING CURRENT
(AC)



DC Current



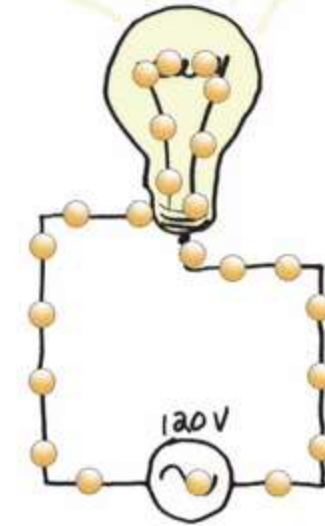
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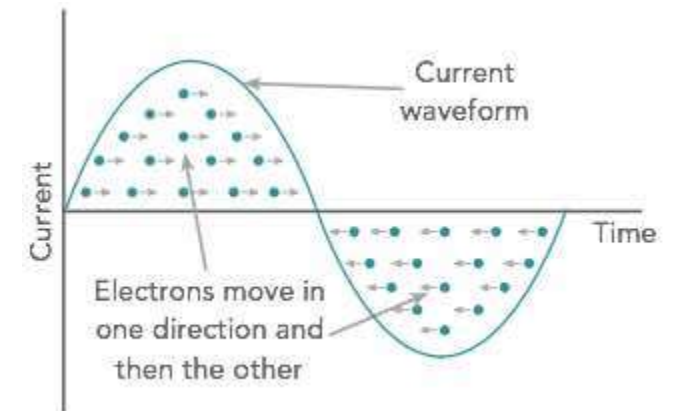
(Conventionally, direction of current is shown from positive to negative)

AC Current

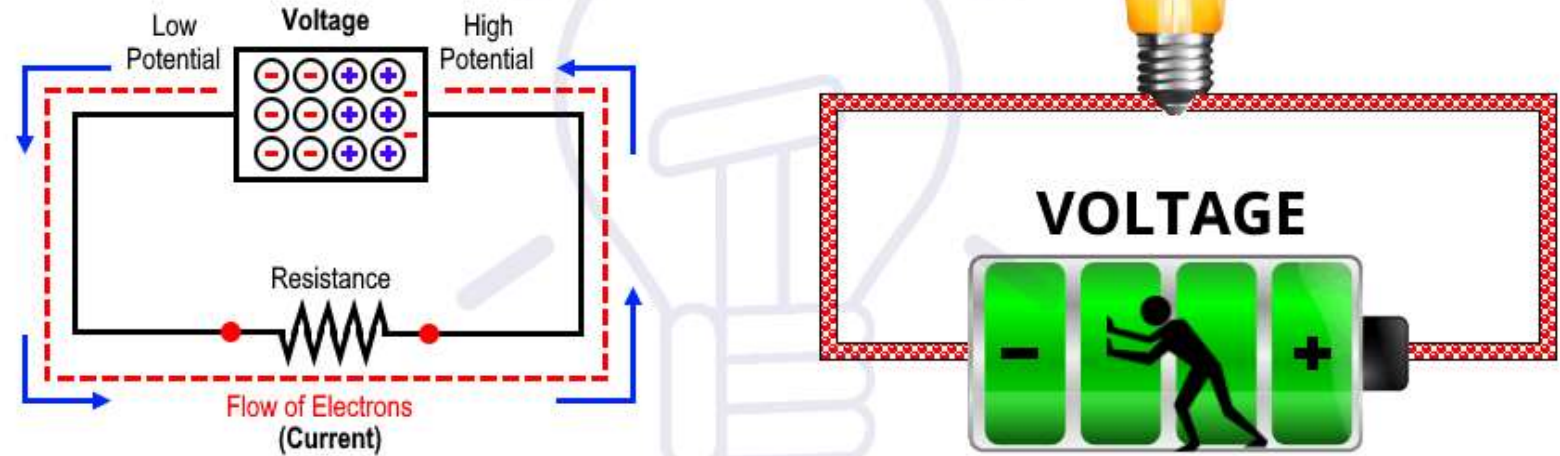
AddOhms



AC
Alternating
Current



Voltage

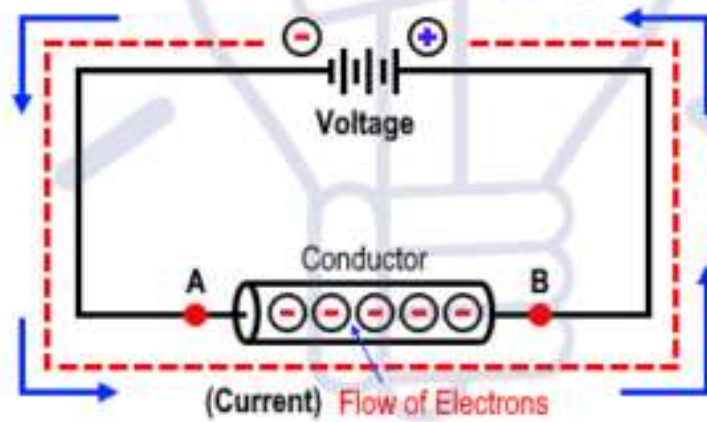
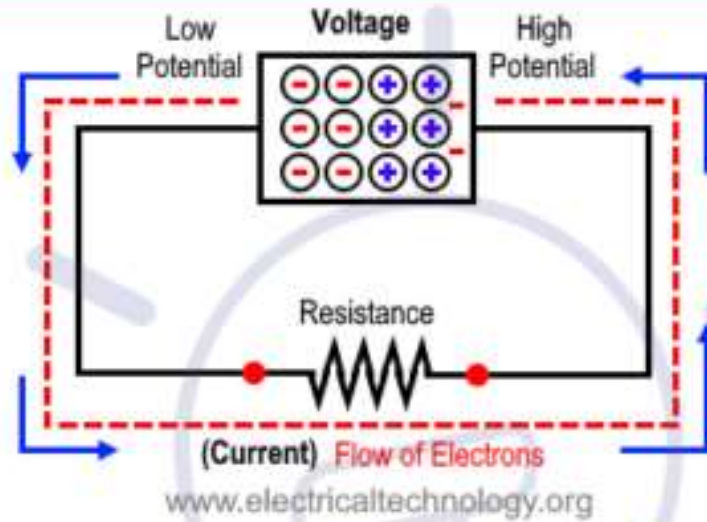


- In electrical circuit, voltage is the force or pressure that is responsible for pushing the charge in a closed looped conductor. The flowing of charge is called current. The voltage is the electric potential between two points; the greater the voltage the greater will be the current flow through that point. It is denoted by letter V or E (used for representing electromotive force).
- The voltage V between two points a and b in an electric circuit is the energy (or work) needed to move a unit charge from a to b; This emf is also known as **voltage or potential difference**. mathematically,

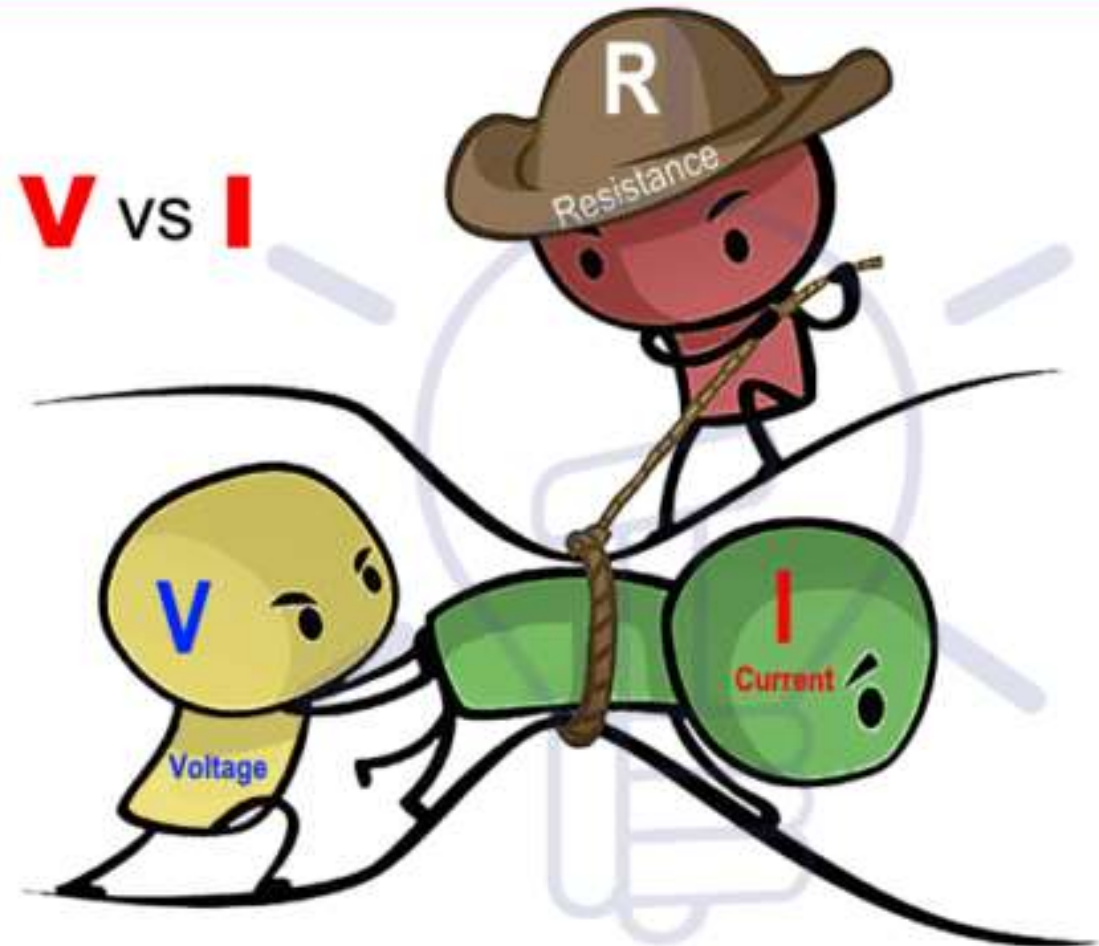
$$v_{ab} = \frac{dw}{dq}$$

Where w is the energy in Joules (J) and q is the charge in coulombs (C)

Difference between CURRENT & VOLTAGE



V vs **I**



Energy

- Energy is the capacity to do work, measured in Joules(J)

The electric power utility companies measure energy in watt-hours (Wh),
where

$$1 \text{ Wh} = 3,600 \text{ J}$$

POWER

- Power is the time rate of expending or absorbing energy, measured in Watts(W).
- Power is the rate of energy consumption, energy can directly be calculated using

$$p = \frac{dw}{dt}$$

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$

$$p = vi$$

Power and Energy

- For practical purposes, we need to know how much power an electric device can handle.
- We all know from experience that a 100-watt bulb gives more light than a 60-watt bulb.
- We also know that when we pay our bills to the electric utility companies, we are paying for the electric energy consumed over a certain period of time.
- Thus power and energy calculations are important in circuit analysis.

Cost of Energy Calculation

- **Estimating Electricity Usage**
- Every appliance or electronic device adds a little something to your bill.
- By figuring out what the biggest energy hogs are in your home ????
- Every change you make = Energy expenses.

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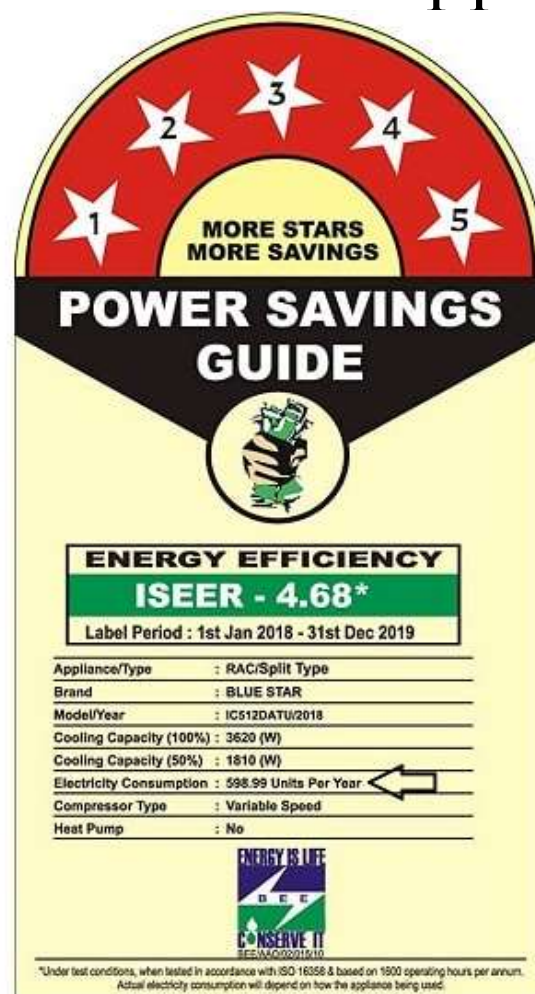
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Cost of Energy Calculation

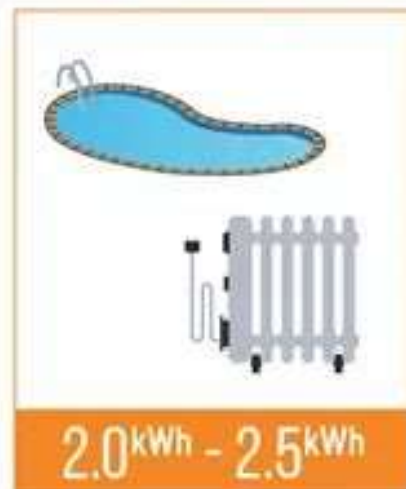
- Calculating the energy cost of an appliance or electronic device .



TYPICAL ELECTRICITY USAGE PER APPLIANCE FOR AN HOUR

Switching off your geyser, air-con/heater, pool pump and other unnecessary appliances, especially during peak times can reduce demand. This can lower the risk of load shedding.

USING YOUR APPLIANCES EFFECTIVELY CAN LOWER THE RISK OF LOAD SHEDDING



Calculate Electricity Consumption - 4 Easy Steps

- Step 1 : Watt-hours(Wh) Per Day

Example : 125 Watts television is running for 3 hours per day

Energy Consumed : $125 \text{ watts} \times 3 \text{ hours} = 375 \text{ watt-hours per day}$

- Step 2 : Convert to Kilowatts

Electricity is measure in kilowatt hours on your electricity bill

1 kilowatt is equal to 1,000 watts .

calculating how many kWh a particular device uses is as easy as dividing by 1,000.

$375 \text{ watt-hours per day} / 1000 = 0.375 \text{ kWh per day}$

1 unit = 1 Kilowatt-Hour (kWh) of electricity.

Calculate Electricity Consumption - 4 Easy Steps

- Step 3 : Usage Over a Month Period
375 watt-hours per day X 30 days =
11.25 kWh per month
- Step 4 : Figuring Out the Cost

Condition	Slab Low	Slab High	Rate (in Rs)
Last Year Consumption ≤ 900 units	1	50	1.45
	51	100	2.6
	101	200	3.6
	201	above	6.9
Last Year Consumption > 900 units and ≤ 2700	1	100	2.6
	101	200	3.6
	201	300	6.9
	301	above	7.75
Last Year Consumption > 2700	1	50	2.68
	51	100	3.35
	101	200	5.42
	201	300	7.11
	301	400	7.98

Calculate Electricity Consumption - Example

- Your refrigerator runs 24 hours a day.
- Most refrigerators consume 300-780 watts of electricity.
- Let's assume you bought a model that uses only 300 watts.

$300 \text{ watts} \times 24 \text{ hours} = 7,200 \text{ watt-hours per day}$

$7,200 \text{ watt-hours per day} / 1000 = 7.2 \text{ kWh per day}$

$7.2 \text{ kWh per day} \times 30 \text{ days} = 216 \text{ kWh per month}$

Common Wattages for Household Appliances

Coffee maker	900-1200 watts
Microwave	750-1100 watts
Toaster	800-1400 watts
Dishwasher	1200-2400 watts
Washer	350-500 watts
Dryer	1800-5000 watts
Iron	100-1800 watts
Ceiling fan	65-175 watts
Space heater (40gal)	4500-5500 watts
Hair dryer	1200-1875 watts
Laptop	50 watts
Computer monitor	150 watts
Computer tower	120 watts
Television 19"-36"	65-133 watts
Television 53"-61"	170 watts

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Assignment-1:(submit ppt)-20 M

- Find out the Wattage of Appliances/Electronic/Electrical Devices in your home.(one appliance per page)
- Find how much power is consumed by all appliances/electronic devices in your home based on their usage per day , find out power consumption in watthours per day and also in units per day. Calculate the same for 30 days. The total no.of units should match to your house current bill units which need to be attached as proof in ppt. Find the total cost per month as per below rate card.

- For 1 to 200 units consumption = Cost 2/- per unit
- For 201 to 500 units consumption = Cost 3/- per unit
- For 501 to 1000 units consumption = Cost 4/- per unit
- For 1001 to any units consumption = Cost 5/- per unit

Note: Place all pictures and calculation in ppt, convert that into pdf , submit in assignment link. Your name and roll no. should be in first page.